# Number 4 — Compound percentages · Inverse percentages

Calculator allowed. Give money to the nearest cent and other answers to 3 s.f. Marks in brackets. ★ = harder question.

**1** **a** Increase €1500 by 12%. [4]

**b** Decrease €1500 by 12%.

**2** A painting is bought for €40 000. It gains 8% in value in the first year and a further 5% in the second year. Find its value after the two increases. [3]

**3** A flat is valued at €800 000. It loses 15% of its value one year and a further 3% the next. Find its value after the two decreases. [3]

**4** A footballer is valued at €24 million. After a bad season his value depreciates by 20%; after the next, good, season it appreciates by 20%. Find his value after the good season. Is he back to €24 million? [3]

**5** A diamond is sold for $38 million. If it appreciates at 7% p.a. for the next 5 years, what is its value after 5 years? [4]

**6** A cheetah's top speed is 110 km/h. Suppose it loses 7% of its top speed each year. Find its top speed after 5 years. [4]

**7** **a** A sun-hat is sold at 25% off for €18. Find its original price. [4]

**b** A drone is sold for €2100, making a profit of 25%. Find the price the seller paid for it.

**8** A sculpture is valued at €$x$. It appreciates by 2% p.a. for 2 years, then depreciates by 3% p.a. for 3 years, and is then sold for €4 million. Find $x$, to the nearest €1000. [4]

**Total: 29 marks**

Check: $n$ years at $R\%$ → multiply by $\left(1 \pm \frac{R}{100}\right)^n$ · successive changes multiply · to find an *original* amount, divide by the multiplier · a 20% fall then a 20% rise is not a return to the start.